Graphs in Machine Learning

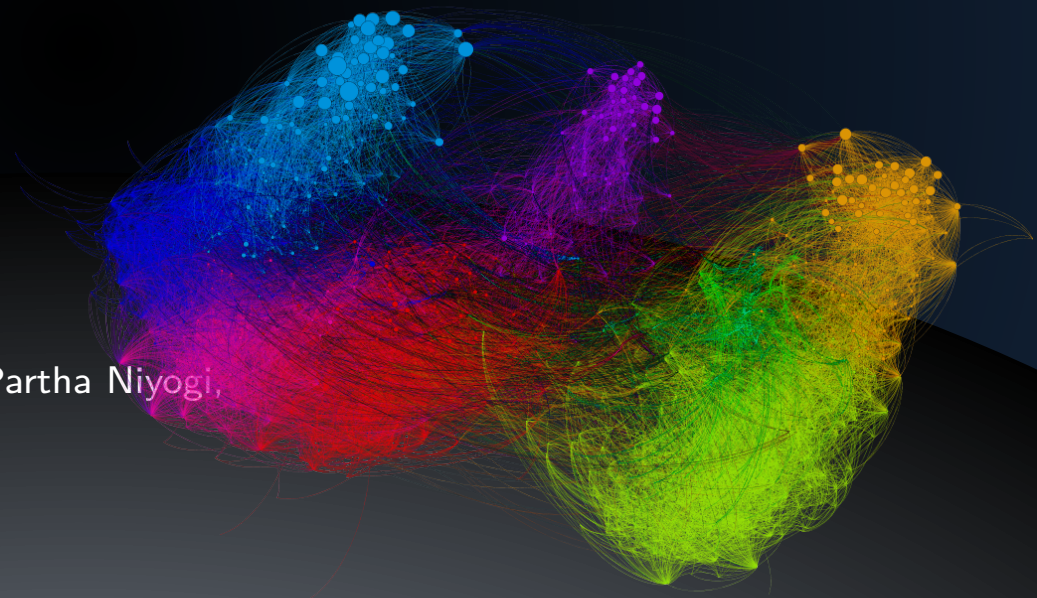
Semi-Supervised Learning Introduction

Why and When SSL Helps

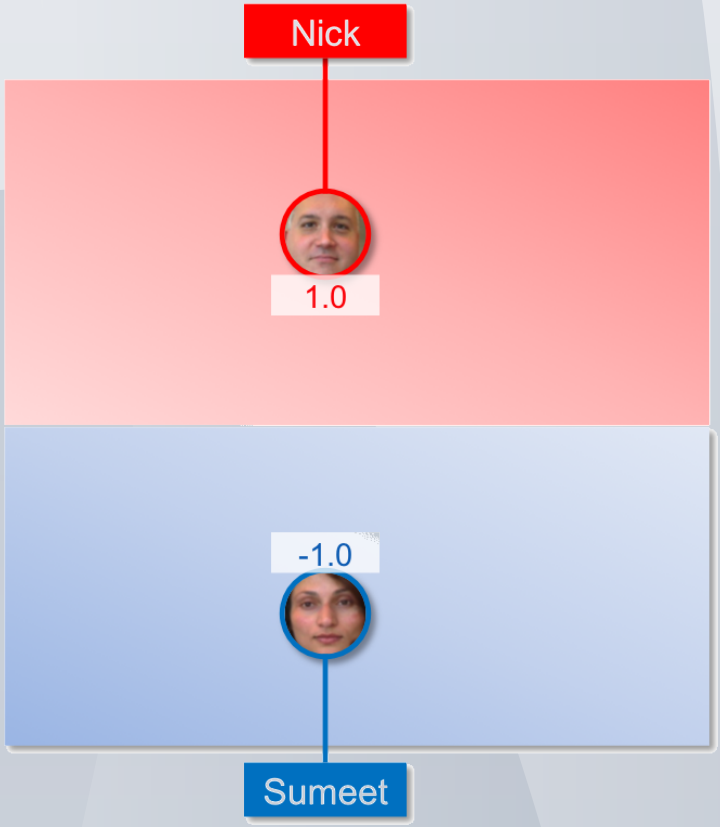
Michal Valko

Isara Labs

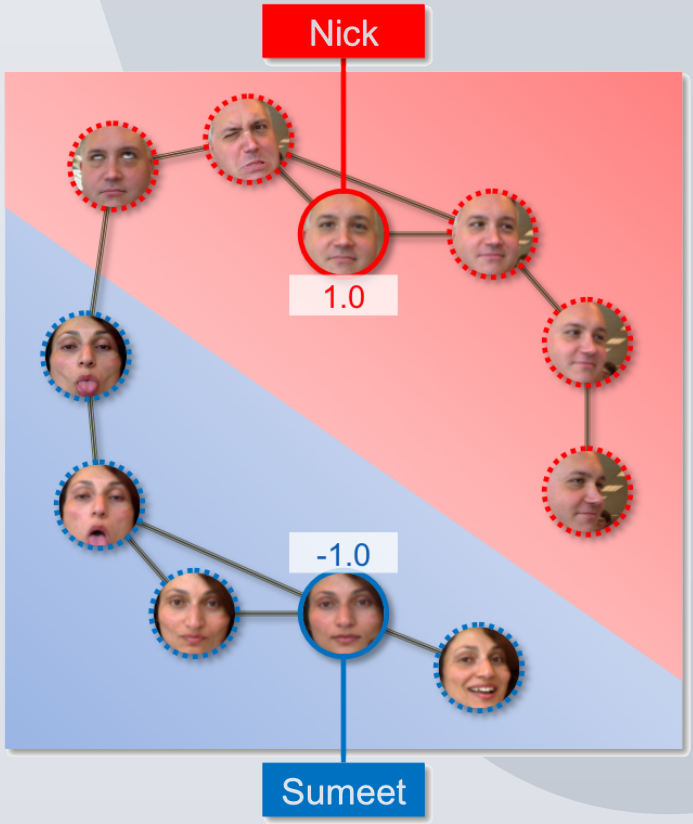
Partially based on material by: Mikhail Belkin, Partha Niyogi, Olivier Chapelle, Bernhard Schölkopf



Semi-supervised learning: How is it possible?

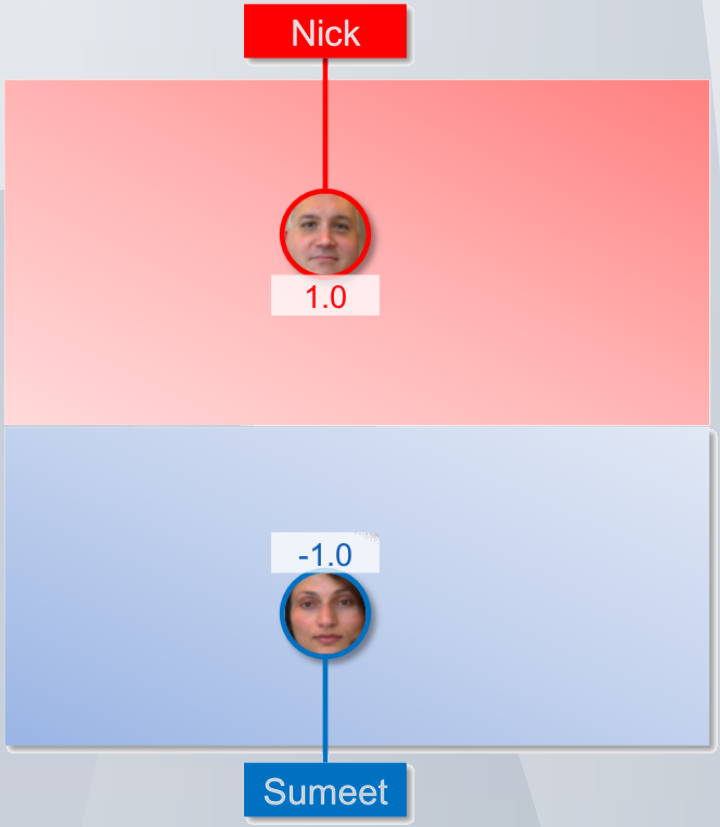


SUPERVISED

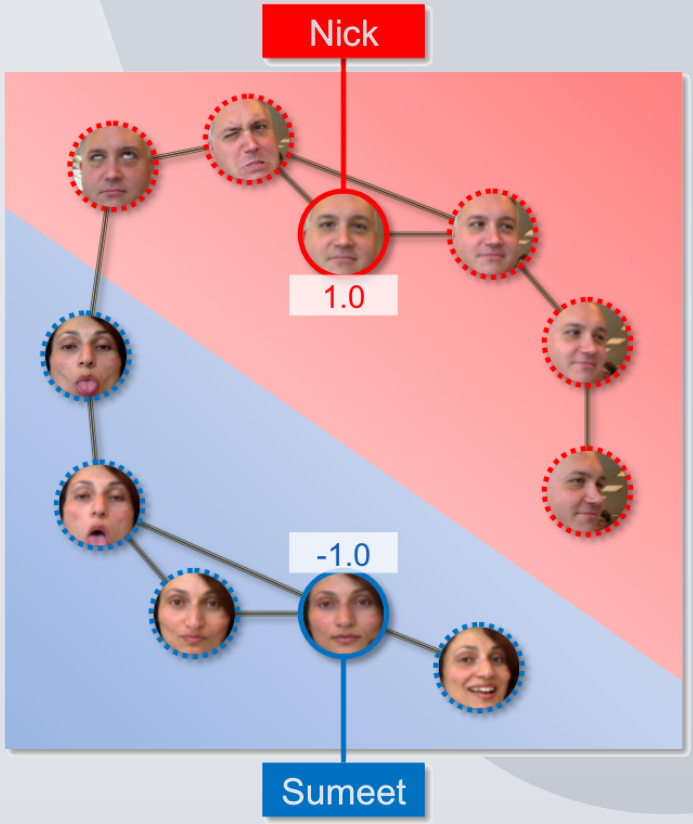


SEMI-SUPERVISED

Semi-supervised learning: How is it possible?



SUPERVISED



SEMI-SUPERVISED

This is how children learn! hypothesis

Semi-supervised learning (SSL)

SSL problem: definition

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Given $\{\mathbf{x}_i\}_{i=1}^{N}$ from $\mathbb{R}^d$ and $\{y_i\}_{i=1}^{n_l}$, with $n_l \ll N$, find $\{y_i\}_{i=n_l+1}^{N}$ (**transductive**) or find f predicting y well beyond that (**inductive**).

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- inductive or transductive/out-of-sample extension

http://olivier.chapelle.cc/ssl-book/discussion.pdf

SSL: Overview: Self-Training

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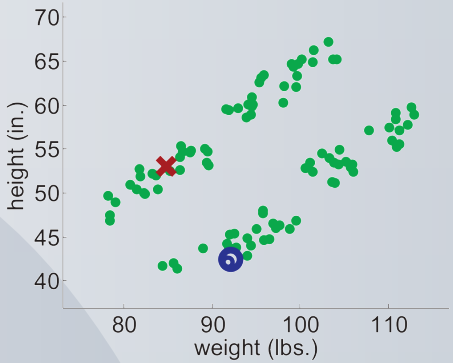
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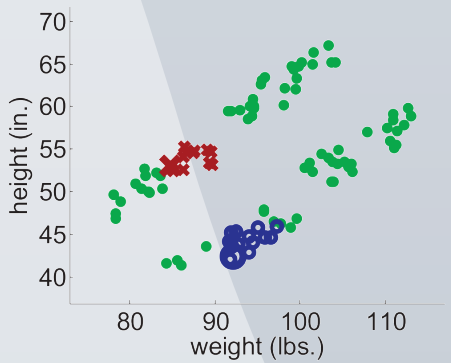
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- its a wrapper method
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- errors propagate (unless the clusters are well separated)

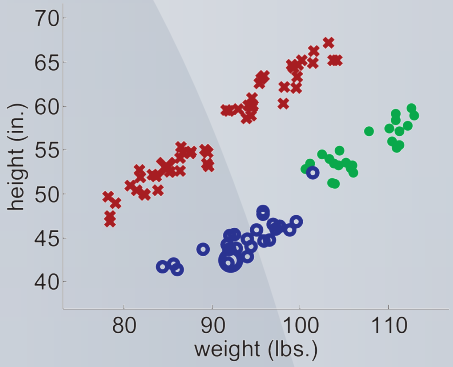
SSL: Self-Training (Good Case)



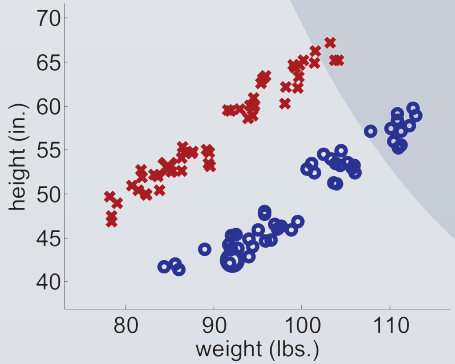
(a) Iteration 1



(b) Iteration 25



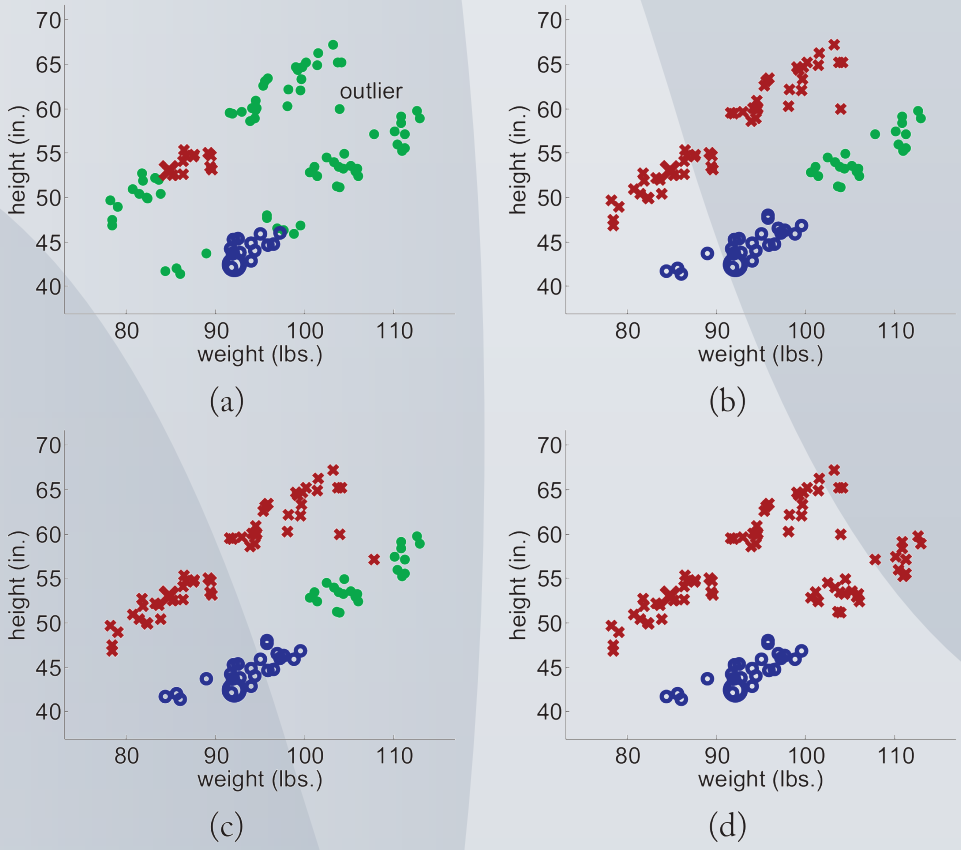
(c) Iteration 74



(d) Final labeling of all instances

Source: **chapelle2006semi-supervised<empty citation>**

SSL: Self-Training (Bad Case)



Source: **chapelle2006semi-supervised<empty citation>**

SSL($\mathcal{G}$)

semi-supervised learning with graphs and harmonic functions

...our running example for learning with graphs

SSL with Graphs: Prehistory

Blum/Chawla: Learning from Labeled and Unlabeled Data using Graph Mincuts
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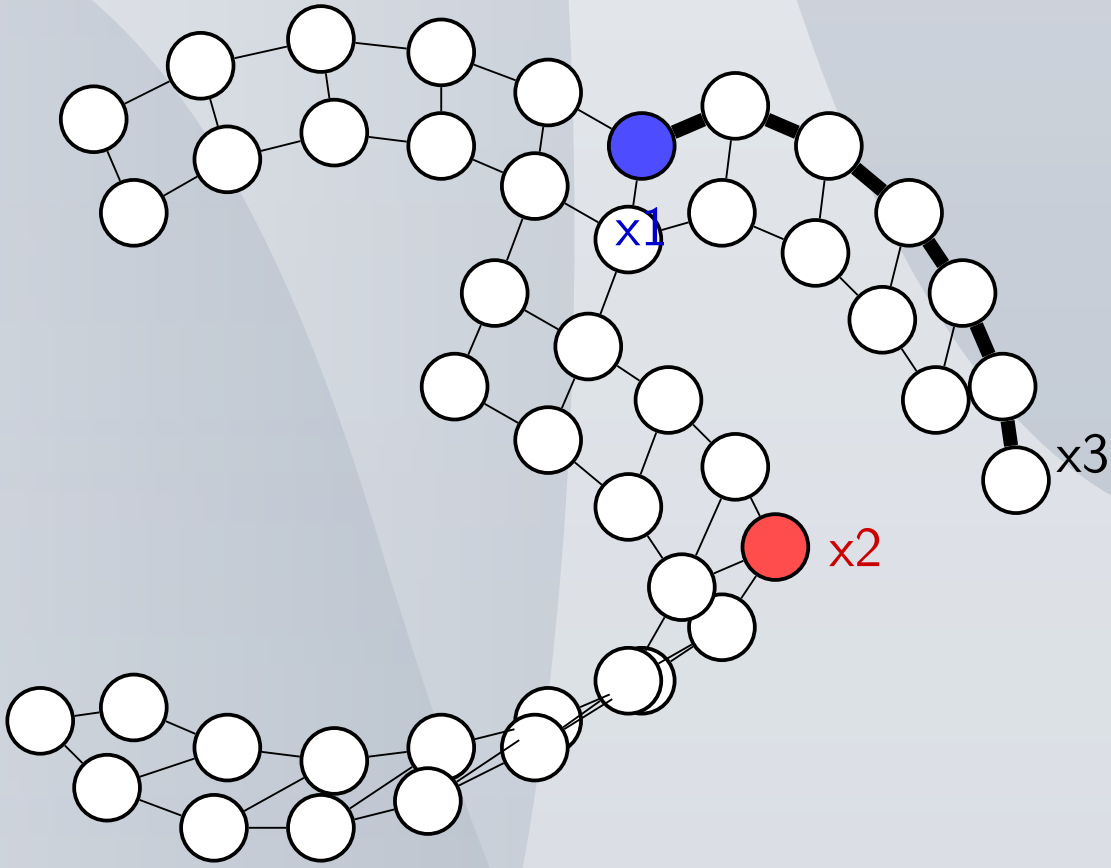
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SSL with Graphs: Prehistory

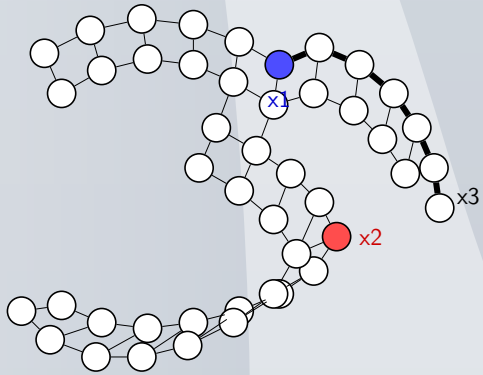
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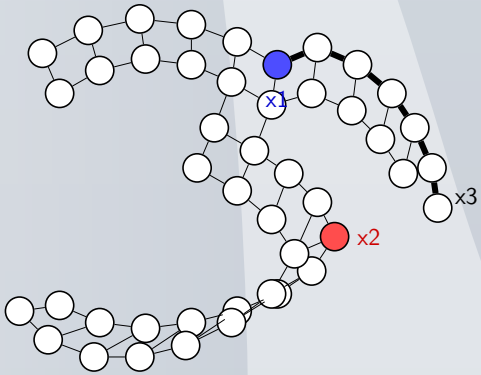


SSL with Graphs: MinCut



MinCut SSL: an idea similar to MinCut clustering

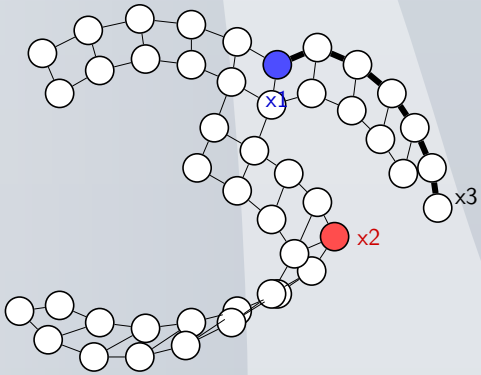
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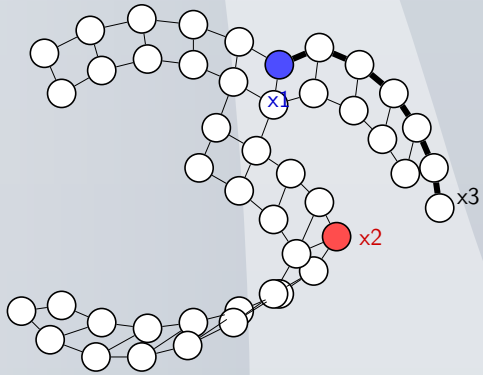
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Where is the link? connected classes, not necessarily compact

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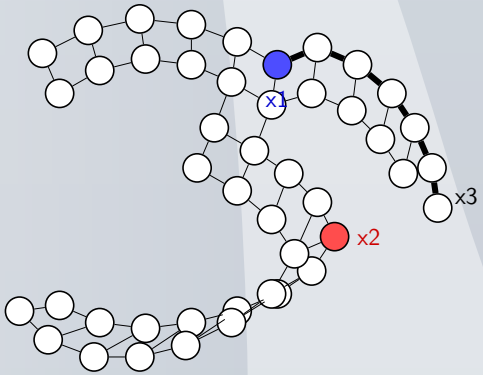


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SSL with Graphs: MinCut



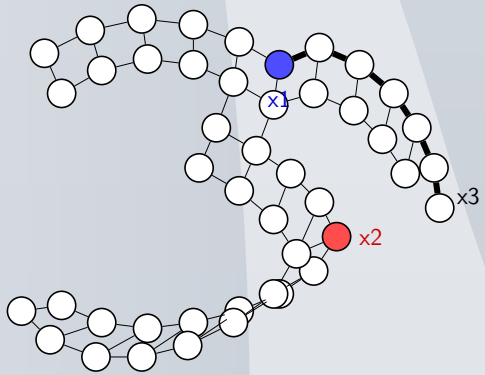
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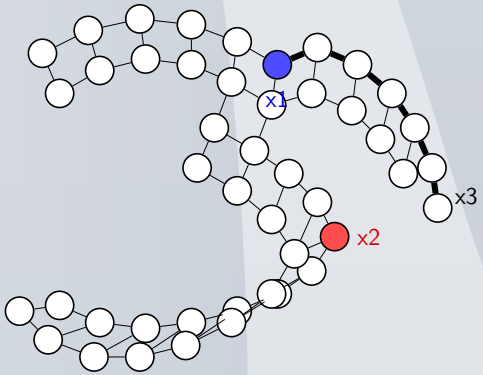
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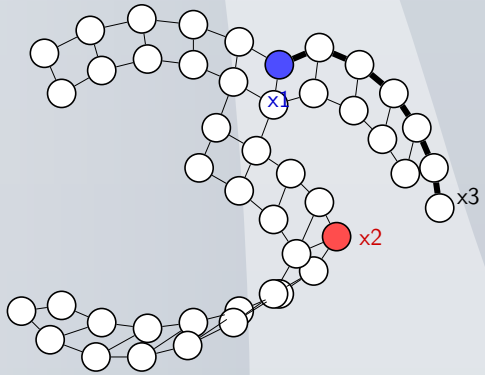
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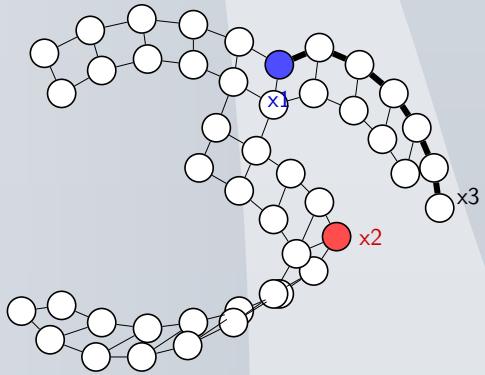
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SSL with Graphs: MinCut

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SSL with Graphs: MinCut

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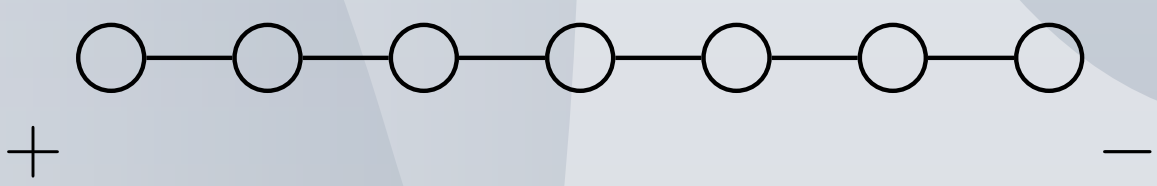
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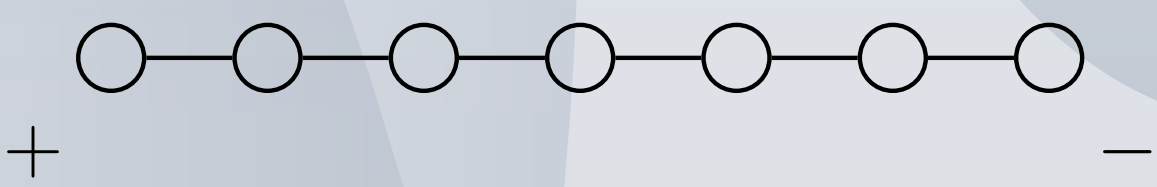
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There are six solutions.

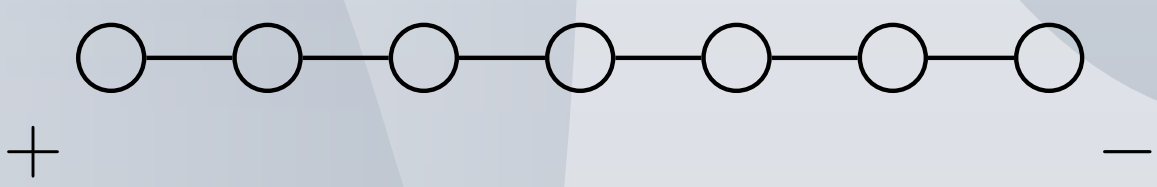
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Can we solve it? It still just MinCut. Are we happy?



There are six solutions. All equivalent.

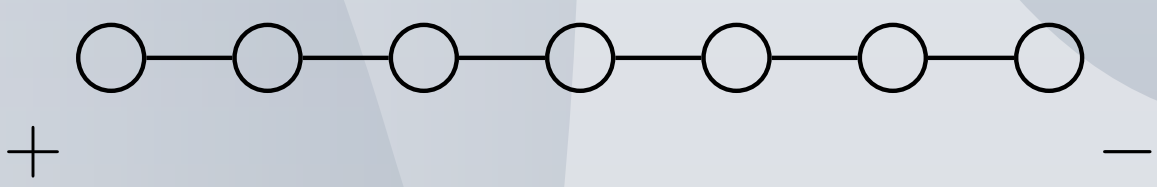
SSL with Graphs: MinCut

Final objective function:

$$\min_{\mathbf{f}\in\{\pm 1\}^{n_l+n_u}} \infty \sum_{i=1}^{n_l} \left(f(\mathbf{x}_i) - y_i\right)^2 + \lambda \sum_{i,j=1}^{n_l+n_u} w_{ij} \left(f(\mathbf{x}_i) - f(\mathbf{x}_j)\right)^2$$

This is an integer program :(

Can we solve it? It still just MinCut. Are we happy?



There are six solutions. All equivalent.

We need a better way to reflect the confidence.

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https://misovalko.github.io/mva-ml-graphs.html